

CURRENCY DETECTION MODEL

By Bipin Subedi

INTRODUCTION

- It is a multi-class classification model to detect the currency of a note image based on the Banknote Net dataset.
- It uses pretrained MobileNetV2 CNN model to embed the image into 256-dimensional vector embeddings.



DATASET USED

BANKNOTE NET

The Banknote_net dataset is a dataset provided by microsoft that contains 256-dimensional vector embeddings of currency images. It has 2 labels currency and denomination and only currency is used for creation of this model, which has 17 classes.



GOALS

GOAL 01

To create a multi-classification deep learning model to classify the currency from the vector embeddings.

GOAL 02

To embed and preprocess raw images into 256-dimensional vector embeddings using banknote_net_encoder model based on MobileNetV2 CNN model.

MODEL ARCHITECTURE

BANKNOTE_NET_ENCODER

Input → MobileNetV2 → GlobalAveragePooling2d → Output

Activation: ReLU

CLASSIFIER

Input (input_size) → Linear (192) → BatchNormald (192) → Linear (96)

→ BatchNormald (96) → Linear (48) → BatchNormald (48) → Linear

(output_size) → Output

Activation: ReLU



USAGE

TRAINING

Use the vector embeddings data to train and validate the classifier model in batches and epochs, and save the best model state.

INFERENCE

Create a Streamlit webapp to take image from user, and use the pretrained encoder model to encode the image, and predict the currency using the classifier model.

MODEL TRAINING

LOSS FUNCTION

CrossEntropyLoss

DROPOUT RATE

0.3

OPTIMIZER

Adam

LEARNING RATE

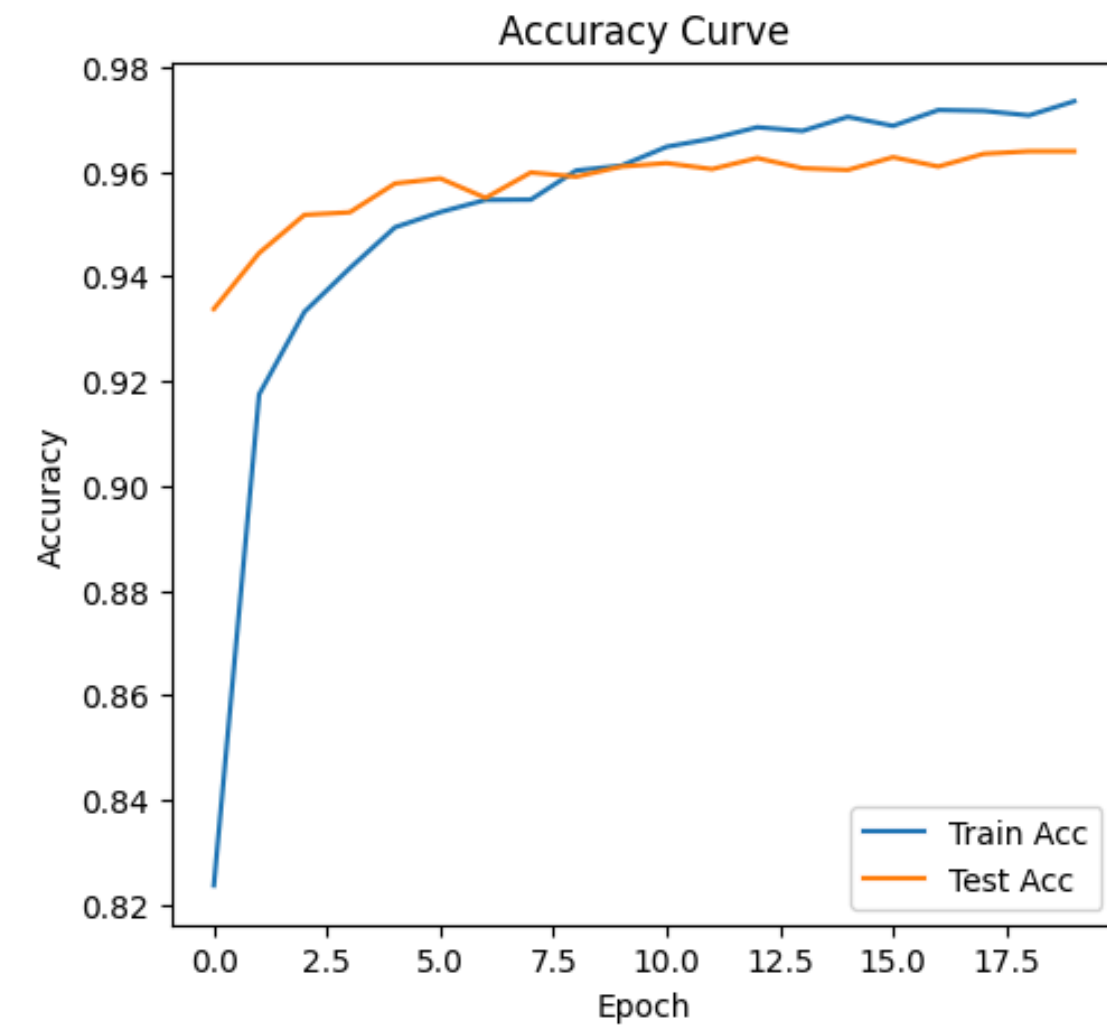
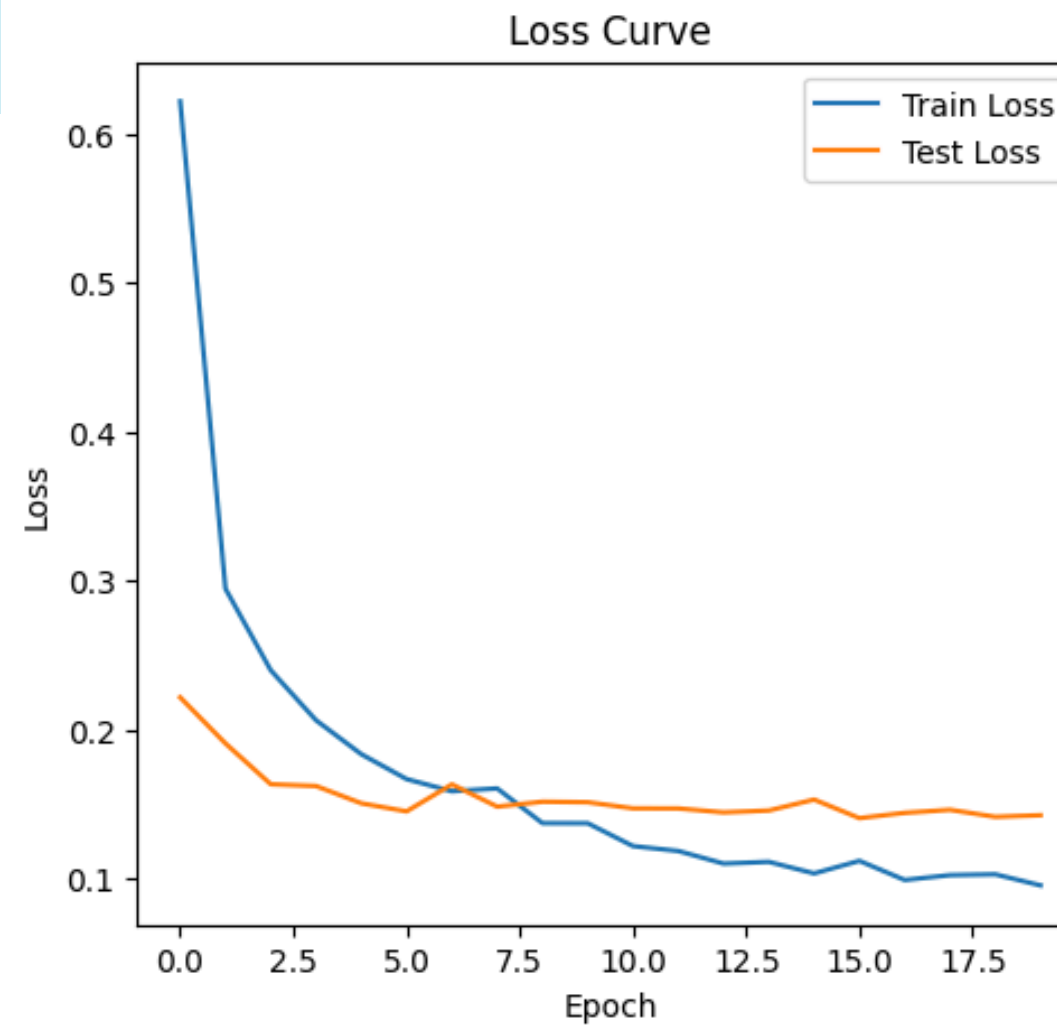
0.01

EPOCHS

20

BATCH SIZE

128



MODEL INFERENCE

- Used Streamlit to generate inference webapp.
- Takes image file input.
- Shows predicted currency with confidence score.

Currency Recognition App

Upload a banknote image to predict its currency type and confidence score.

Choose a banknote image...



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG, WEBP

Browse files



usd.jpg 273.6KB



Predicted Currency: **USD**

Confidence Score: **100.00%**

The background features a light gray gradient with abstract teal geometric elements. In the top-left and bottom-left corners, there are teal triangles and lines forming rectangular frames. In the top-right and bottom-right corners, there are teal triangles and lines, with a 4x5 grid of small teal circles in the top-right and bottom-left. The text is centered in the middle of the image.

THANK YOU

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